课堂讲稿与方法说明
Class talk and methods

一页 DEM 主展示 + 单独 PUDO 备用页。此说明册不代替老师要求的一页展示。

Use the DEM slide as the main class page. Keep PUDO as a separate backup. The paper and this booklet are supporting material, not extra pages to present.

课堂主要展示 DEM 那一页。PUDO 独立备用，不能把两套数据混在同一风险率中。长论文和本说明册用于回答追问。

老师要求 / What the instructor requested

要求 / Requirement	我们如何完成 / Our delivery
读记录与叙述 / Read data and narratives	重点危害、原文来源与证据强度。
估计频率与后果 / Estimate frequency and impact	184 个覆盖日的报告率；每个后果等级对应自己的事件计数。
风险矩阵 / Risk matrix	使用课堂 5×5 标尺，显示 H2 含推断与严格证据两种结果。
一页、清楚、可解释 / One clear slide	主结论 + 矩阵 + 数字 + 限制；不要讲全部数学模型。

我们最重要的贡献 / Our key contribution

We checked what the source can actually support. The biggest issue is not a difficult equation: it is whether the event counted in the probability matches the consequence used in the matrix.

我们核对原文能支持到哪一步。关键不是公式难，而是“算概率的事件”和“放在横轴上的后果”是不是同一类事件。日志时长也不能直接写成救护车延误。

Source / 依据: Instructor class transcript and September 28 email supplied by the user; Jonathan source revision 06fdb2e9; independent numerical checks, 2026-10-05. Speaker roles are suggestions, not an agreed meeting plan.

DEM 主展示：简单英文稿
DEM: simple spoken script

约 90-120 秒；可按需要删去中间一段。中文是理解辅助，不需要念。

We looked at reports of self-driving cars blocking traffic in San Francisco. We used 117 reports over 184 covered days. This is about 19 reports per 30 days. The middle duration was 24 minutes.

我们研究旧金山自动驾驶车辆阻挡交通的报告。频率窗口中有 117 条报告、184 个覆盖日，约每 30 天 19 条。时长中位数为 24 分钟。

We used the class risk chart to compare hazards. Collisions, emergency access, and long single-car blockages are the main concerns. The chance is for a whole group over 30 days, not for one trip or one person.

我们使用课堂风险矩阵比较危害，重点是碰撞、应急通行及长时间单车堵塞。概率针对这一类事件在 30 天内至少发生一次，不是某一趟或某个人的风险。

Emergency access needs care. Some reports suggest a problem at an emergency scene, but do not clearly say responders were blocked. With these reports included, the rating is Very high. With only clear evidence, it is Medium. We show both.

应急通行需要谨慎。一些记录有应急场景，却没有明确写响应人员被阻挡。包含这些推断记录时评级为 Very high；仅明确证据时为 Medium。我们同时显示两种结果。

About 29 percent of reports lasted an hour or more. They account for 74 percent of logged event time. This supports faster recovery as a priority. But logged time is not medical delay or measured harm.

约 29% 的报告持续至少一小时，贡献了 74% 的日志时间。这支持把快速恢复作为重点，但日志时间不是医疗延误或实测伤害。

你可独立补充的一句 / A short part you can say

My part was checking the original cases and the numbers. I kept clear evidence separate from uncertain details, especially for emergency access.

我的部分是核对原始案例与数字，特别把应急通行的明确证据和不确定信息分开。

Source / 依据: Slide 1; independently verified cohort and durations. Risk rubric and base placements are Jonathan’s method, not newly invented here.

PUDO 备用稿与常见追问
PUDO backup and short answers

For PUDO, we compared the last quarter with the previous three quarters together. The last rate was 1.87 complaints per 100,000 trips. The earlier rate was 1.71. The rate ratio was 1.09, but the interval was wide. We did not find clear evidence of a rate difference. This does not prove the rates are equal.

PUDO 比较最后一季与前三季合并。最后一季每十万趟 1.87 条投诉，之前 1.71 条。率比 1.09，但区间较宽，未发现明确差异；这不证明相等。

问 / Question	简单回答 / Simple answer
Is 41% the chance for one person? 41% 是一个人的概率吗？	No. It is the chance of at least one qualifying report in the whole group over 30 days. 不是，是 30 天内整类至少出现一条符合条件报告的概率。
Did the ambulance lose 176 minutes? 救护车耽误 176 分钟吗？	We do not know. 176 minutes is the logged event duration. 不知道；176 分钟是事件日志时长。
Do you know the injury risk per trip? 知道每趟伤害风险吗？	No. DEM has no trip or mileage denominator and no measured harm outcome. 不知道；DEM 没有趟数/里程分母，也没有实测伤害目标。
Why did the new model not help? 新增模型为什么没提升？	These fields and this small sample did not predict long events well in our test. 这次小样本验证中，这些字段对长事件预测不好。

Source / 依据: Instructor PUDO workbook: ME 575 Class Final Data - PUDO - Waymo.xlsx, Sheet1!A7:D10. Latest 79 / 4,220,075 trips; prior 177 / 10,330,467 trips. Exact conditional two-sided p=0.535; 95% rate-ratio CI=0.827-1.432.

H2：先把问题问对
H2: count the right event

H2 = emergency-response obstruction / 应急响应通行受阻。Severe 是潜在后果标尺，不是已发生死亡或重伤。

口径 / Definition	计数与未来 30 天概率 Count and 30-day probability	放置 / Placement
全部应急类别 All H2, Major or worse	6 条 / 6 reports 62.5%	Likely × Major Very high
危及生命应急场景 Life-threatening scene, inferred obstruction	3 条 / 3 reports 41.1%	Moderate × Severe Very high
仅明确描述受阻 Explicit obstruction only	1 条 / 1 report 20.3%	Unlikely × Major Medium

The 41% is not 3 divided by 6. We saw 3 qualifying events in 184 days. A count model estimates the chance of at least one similar report in a new 30-day period.

41% 不是 3÷6。我们在 184 天观察到 3 条符合该后果层级的记录；计数模型估计另一个类似条件的 30 天内至少出现一条的概率。

Do not combine all 6 emergency reports with the worst Severe consequence. That would count the common event but attach the rarer consequence, pushing the matrix to Extreme.

不能把全部 6 条应急报告的概率配到最坏的 Severe 后果上。那是在用更常见事件的概率搭配更稀少的后果，会把矩阵推成 Extreme。

INC-002 says medics changed their approach near blocked traffic. It supports an access concern. It does not measure 176 minutes of medical delay or prove the AV caused the patient’s illness. It is outside the main frequency window because it is a February record.

INC-002 写医护人员在堵塞交通附近改变进场方向，支持通行问题。但没有测量 176 分钟的医疗延误，也不证明 AV 导致病人患病。该条为二月记录，不在主频率窗口内。

Source / 依据: Jonathan risk_matrix.py and evidence_flags.csv at 06fdb2e9; independent numerical_tests.py: Severe IDs INC-064/089/102; explicit Major ID INC-079. INC-002: primary workbook DEM Incidents, physical row 3.

为什么选这些模型？
Why these models?

先决定问题和目标，再决定模型。不是模型越多越好。

要回答什么 / Question	适合的方法 / Method	边界 / What it cannot show
把原文放进哪类危害？ Which hazard fits the text?	规则、文本分类器 Rules / text classifiers	与参考标签一致，不等于事实已证实。 Agreement is not factual truth.
未来 30 天会出现多少报告？ How many reports in 30 days?	Poisson 计数 + Gamma 后验 Poisson counts + Gamma posterior	假设覆盖可比、报告率相对稳定；不是每趟伤害率。
原文后果有多严重？ How serious is the consequence?	预先定义的影响标尺 + 原文复核 Rubric + source review	不能用一条关键词证明伤害已经发生。
哪些事件会超过一小时？ Which events last an hour?	逻辑回归 / 浅层分类树 Logistic / shallow tree	目标为 ≥60 分钟；样本外验证才有意义。
最后一期投诉率不同吗？ Is the last complaint rate different?	条件精确率比检验 Exact conditional rate-ratio test	分母用 trips，不是 Student t test。
怎样把时间分组？ How can time be grouped?	K-means / 聚类是可选工具	分组规则和 K 是选择；不是二分类必需步骤。

课堂说法需要修正的地方 / Important corrections

Linear regression does not require the raw target or every predictor to be normally distributed. Normality concerns the errors for standard small-sample inference. Logistic regression can use continuous and categorical predictors. Two sibling tree leaves can both say Yes without being invalid.

线性回归不要求原始目标或所有输入都正态；常见小样本推断的正态假设针对误差。逻辑回归可以用连续和类别输入。决策树两个子叶都预测 Yes，并不能单凭这点判定树无效。

Source / 依据: Statistical interpretation, not a restatement of every lecture claim. New prediction design uses supplied nominal areas as categories and clock sine/cosine, not arbitrary ordered region numbers.

Source / 依据: Primary technical references: scikit-learn Linear Models and Decision Trees documentation (scikit-learn.org/stable/modules/linear_model.html and /tree.html); NIST e-Handbook, section 4.2.1.4, random-error normality (www.itl.nist.gov/div898/handbook/pmd/section2/pmd214.htm).

Jonathan 做了什么？
What Jonathan built

步骤 / Step	产出 / Output	我们怎样检查 / Our check
1 数据与范围 / Data and scope	原始 123 → 范围内 120 → 窗口内 117	核对排除 ID、日历覆盖、字段和行号。
2 文本危害分类 / Text hazards	H1-H7、规则、传统模型、LLM 比较	分清标签来源、训练内结果与样本外结果。
3 影响评分 / Impact scoring	取后果、时长、车辆数分数的最大值	分开实测后果与潜在情景；核对阈值。
4 频率 / Frequency	报告率、区间、未来至少一次概率	独立重算 Poisson/Gamma 数字。
5 矩阵 / Matrix	每类按同一后果层级的计数放置	核对 H2 推断与明确证据；不配错事件。
6 敏感性 / Sensitivity	窗口、证据、阈值、标注者变化	说明哪些结论稳健、哪些依赖定义。

A 98% classification score is agreement with a reference labeling scheme, not 98% proof that the reports are true. A second coder’s kappa near 0.95 measures agreement beyond chance; an LLM second coder is not an independent human gold standard.

分类 98% 是与参考标注方案的一致程度，不是“事实真实率 98%”。第二标注者 κ≈0.95 衡量超过随机一致的程度；LLM 第二标注者不是独立人工金标准。

Macro F1 gives each hazard equal weight. Micro F1 pools all decisions; in single-label multiclass classification it equals accuracy. A common class can make accuracy high while a rare emergency class is missed.

Macro F1 对每个危害类别等权。Micro F1 汇总所有判断；单标签多类时等于准确率。常见类别可能把准确率抬高，却仍漏掉稀少的应急类别。

Source / 依据: Jonathan paper and pinned scripts. Existing independent classifier reruns: 20 logistic fits, mean fold macro F1 0.451, accuracy 0.767; H2 recall was zero in every repeat. Narrow to that tested configuration; guided LLM had more detailed guidance than short-description baselines.

新增回检：能预测长事件吗？
New check: can we predict long events?

探索性检查，不是预注册研究；不证明所有模型都无效。

Question: can vehicle count, start clock time and nominal area predict a logged event lasting at least 60 minutes? We excluded duration, clear time, hazard and impact labels from the inputs.

问题：车辆数、开始时刻和名义区域，能否预测日志事件 ≥60 分钟？输入中不使用时长、清除时间、危害标签或影响分数，避免答案泄漏。

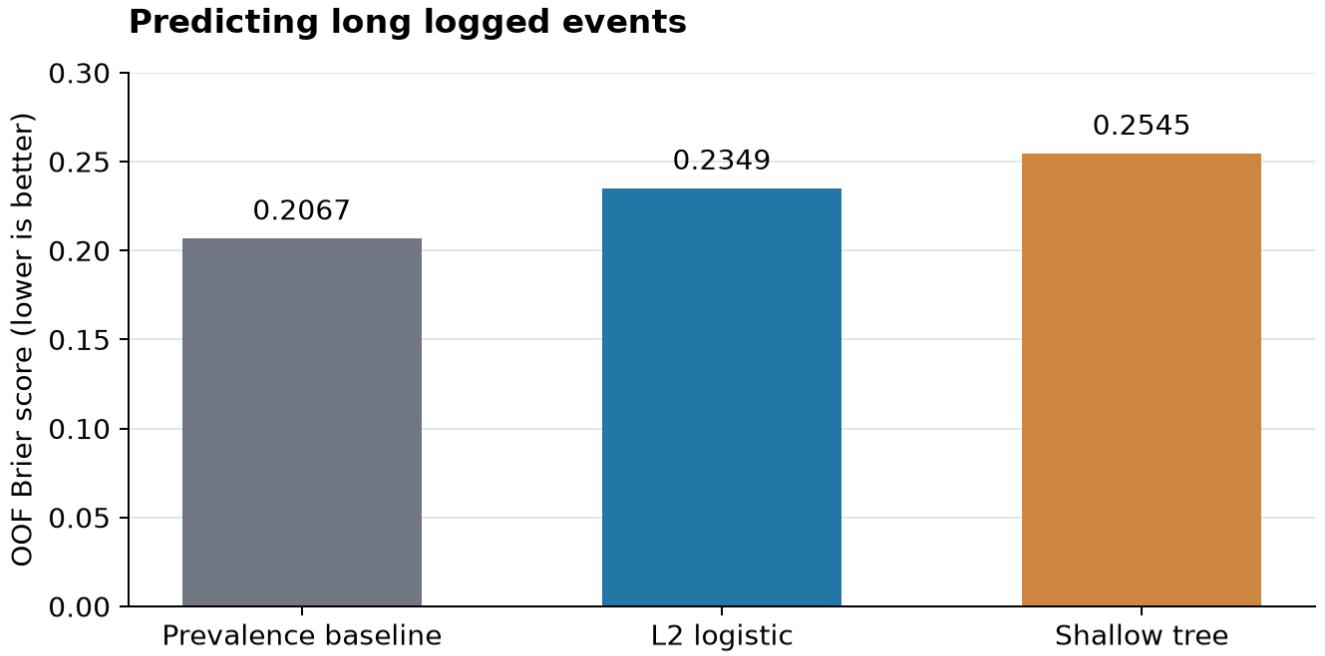
We used the same five date-grouped test folds for all models. All reports from one date stayed together. Imputation and encoding were learned inside the training folds only. There were 117 reports, 34 long events and 80 distinct dates.

三个模型使用相同的五折按日期分组验证，同一天不能拆到训练与测试两边。缺失填补和编码只在训练折拟合。共 117 条、34 条长事件、80 个日期。

模型 / Model	Brier ↓	ROC AUC	Average precision
简单基线 / Baseline	0.207	0.461	0.274
逻辑回归 / Logistic	0.235	0.395	0.295
浅层树 / Shallow tree	0.254	0.448	0.264

Source / 依据: Our model_results.json and 117 saved out-of-fold predictions. Same date-grouped folds for each method; no tuning. Retrospective recorded fields, not an operational early-warning system.

概率误差图：越低越好
Probability error: lower is better



117 events; 34 with logged duration ≥60 minutes. Five folds group calendar dates.
Descriptive exploratory comparison. Both fitted models have greater probability error than the baseline.

Both models had worse probability error than the simple baseline. We did not find convincing predictive improvement. The useful conclusion is to improve data and recovery, not claim these inputs can reliably flag the next long event.

两种模型的概率误差都比简单基线更高，未发现令人信服的预测提升。可讲的结论是改善数据与恢复流程，而不是宣称这些输入能可靠预警下一条长事件。

Source / 依据: Our new duration_model.py, model_results.json and out-of-fold predictions, 2026-10-05. Retrospective/exploratory: recorded vehicle count is not established as available at event start. Brier measures mean squared probability error; lower is better. Fold spread is not a confidence interval. No held-out unseen future month is claimed.

A baseline gives the training-set share of long events to every test report. A more complex model must improve on this simple reference. More computation alone does not prove better prediction.

基线把训练数据中长事件的比例作为每条测试记录的概率。复杂模型至少应胜过这个简单参照；计算更多，不等于预测更好。

Source / 依据: Metric definitions: scikit-learn.org/stable/modules/model_evaluation.html. Model documentation does not establish validation for this dataset.

频率、区间、分布分别问什么？
Rate, interval and distribution

东西 / Quantity	解释 / Meaning
Histogram / 直方图	看时长或其他数值分布：是否偏斜、长尾、有极端值。不能替代报告率的置信区间。
Rate / 报告率	117÷184×30 = 19.1 条 / 30 天；单位按日历时间，不按每趟。
95% confidence interval / 置信区间	频率参数的估计不确定性：15.8-22.9 条 / 30 天。不是下月事件数必落在这里的概率。
95% credible interval / 贝叶斯可信区间	给定模型、先验和数据，未知平均率的后验范围。后验均值 19.16，区间 15.85-22.77。
Predictive interval / 预测区间	未来 30 天事件数：该模型下 95% 后验预测区间 11-29 条。额外包含未来计数随机性。
At least one / 至少一次	某类、某后果等级、未来 30 天至少一条。例如 H2 Severe 3 条 → 41.1%。

Poisson describes counts. Gamma describes uncertainty in the average rate. Bayes updates that uncertainty after seeing the data. The extra 0.5 comes from Jeffreys prior, not from half a new report.

Poisson 描述计数；Gamma 描述未知平均率的不确定性；Bayes 在看到数据后更新。多出来的 0.5 来自 Jeffreys 先验，不是凭空添加半条报告。

The 30-day horizon is any new comparable 30-day period under the model assumptions. It is not automatically November, and the dataset does not prove that future conditions will stay the same.

30 天窗口指模型假设下另一个可比的 30 天，不自动等于某年十一月。数据没有证明未来车队规模、覆盖和报告习惯不变。

Source / 依据: Independent frequency verification; Gamma(shape=n+0.5, rate=T), T=184 days. Predictive probability = 1-[T/(T+30)]^(n+0.5). Calendar-day clustering checks suggest modest extra variability; Poisson intervals may be optimistic.

下一步人工复核与自己的分支
Human review and our own branch

The previous 12 cases were discussed, not a fully blind validation set. We prepared 24 additional cases with old labels hidden. Their review fields are blank. No new human agreement score or revised risk estimate is claimed yet.

此前 12 条已经讨论过，不能称为完全盲法验证。新增 24 条隐藏旧标签，人工填写栏仍为空；目前不声称已完成新增人工一致率或新的风险结果。

顺序 / Step	具体做什么 / Action
1 Independent review / 各自独立看	读英文原文与来源行；填主类别、多标签、explicit/inferred/unknown、已记录/潜在后果。
2 Compare / 对照	先保存各自原判断，再比较分歧；unknown 不自动改成 No。
3 Adjudicate / 共同审定	逐条写依据与最终规则，不仅给一个最终类别。
4 Re-run / 重算	只在共同决定了标签或范围后更新计数、严重等级、概率和矩阵；同时保存旧版。

Our repository is CJW0007/curbrisk-av. The separate audit branch is audit/dem-classroom-20261005. It does not change Jonathan's repository. The older PUDO work and uncommitted files are left intact.

自己的仓库是 CJW0007/curbrisk-av。本次独立分支为 audit/dem-classroom-20261005，不改 Jonathan 的仓库；原来 PUDO 的未提交修改保持原样。

New artifacts and analysis are based on Jonathan's pinned source, but the extra model comparison and numerical re-checks are ours. The public package excludes private messages, meeting transcripts and raw medical narratives.

数据与基础框架引用 Jonathan 的固定版本；新增模型对比和数值回检是我们的。公开包不包含私人短信、会议逐字稿和医疗叙述原文。

Source / 依据: Expanded review packet source row crosswalk; model source manifest; source workbook SHA256 primary cd01c0ec…b60a54, analyzed e18d64c7…edee1. The repository README records full hashes and commands.